Unit 2 - Chapter 1

Machine Learning Approaches in NLP

Easy & Detailed Study Notes

These notes follow the topics in your college PDF and explain them in simpler language, with extra examples and step-by-step explanations for studying and exam preparation.

Topics covered: NLP and Machine Learning, Text Classification, Sentiment Analysis, Spam Detection, Topic Modeling (LDA), Keyword Extraction, Named Entity Recognition (NER), and a final comparison/revision section.

A) Classical NLP Tasks

1. What is Natural Language Processing (NLP)?

Natural Language Processing, or NLP, is a branch of Artificial Intelligence that helps computers work with human language. Human language includes the words we speak, type, read, or write. NLP tries to make computers understand this language well enough to perform useful tasks.

For example, when an email service detects spam, a chatbot understands a question, a search engine understands what you typed, or a system decides whether a review is positive or negative, NLP is being used.

Why is NLP difficult?

Human language is not always exact. The same word can have different meanings, people use slang and spelling mistakes, and the meaning of a sentence often depends on context. A computer therefore needs methods for turning language into a form it can process.

> Simple example
> The word "bank" can mean a financial institution or the side of a river. A good NLP system must use the surrounding words to understand which meaning is intended.

Role of Machine Learning in NLP

Earlier NLP systems depended heavily on manually written rules. For example, a programmer might create a rule saying that an email containing certain words should be marked as spam. This approach becomes difficult because language has too many variations.

Machine Learning (ML) improves NLP by allowing a model to learn patterns from examples. Instead of writing every rule manually, we provide training data and allow the algorithm to discover useful patterns.

- Rule-based approach: Humans write the rules.
- Machine-learning approach: The computer learns patterns from data.
- Main benefit: ML can handle large amounts of text and can often adapt better to real-world language.

A common NLP + ML pipeline

1. Text Data - Collect sentences, reviews, emails, documents, tweets, etc.
2. Preprocessing - Clean and normalize the text.
3. Feature Extraction - Convert text into numbers that a model can process.
4. Train ML Model - Let an algorithm learn patterns from examples.
5. Prediction / Output - Use the trained model on new, unseen text.

1. Text Classification

Definition

Text Classification is the process of automatically assigning a predefined category or label to a piece of text. The possible labels are decided before training the model.

It is called a supervised learning task when the training data already contains the correct labels. The model studies many examples of text together with their labels and learns how to classify new text.

Examples from the notes
Email -> Spam / Not Spam
News -> Sports / Politics / Business
Review -> Positive / Negative
Complaint -> Urgent / Normal

How Text Classification Works - Step by Step

Step 1: Collect the Dataset

A dataset is a collection of examples used to train the model. For text classification, each training example normally contains the text and its correct category.

Sentence	Category
India won the cricket match.	Sports
Stock market reaches new high.	Business
New government policy announced.	Politics

The model learns which words and patterns are commonly connected with each category. For example, words such as cricket, match, player, and tournament may frequently appear in Sports documents.

Step 2: Text Preprocessing

Raw text can contain uppercase letters, punctuation, common words, and different forms of the same word. Preprocessing cleans the text so that the model can focus on useful information.

- Lowercasing: Convert words such as Cricket and CRICKET to cricket.
- Remove punctuation: Remove symbols such as commas, full stops, or unnecessary special characters when they do not help the task.
- Stop-word removal: Common words such as the, is, a, and are may be removed in some classical NLP systems.
- Tokenization: Split text into smaller units, usually words or subwords.
- Stemming: Reduce a word to a rough root form. Example: playing -> play.
- Lemmatization: Reduce a word to its meaningful dictionary base form.

Notes example
Original: The players are playing cricket.
Processed: player play cricket

Step 3: Feature Extraction

A machine-learning algorithm works with numbers, not raw sentences. Therefore, text must be converted into numerical features.

Bag of Words (BoW)

Bag of Words creates a vocabulary of words and represents a document using the presence or frequency of those words. It usually ignores word order.

> Example
> Sentence: "I love NLP"
> Vocabulary: I, Love, NLP, AI
> Vector: [1, 1, 1, 0]
>
> The first three values are 1 because those words appear. AI receives 0 because it does not appear.

Limitation: BoW does not naturally understand meaning or word order. For example, two sentences with similar words may receive similar representations even if their meanings differ.

TF-IDF

TF-IDF stands for Term Frequency-Inverse Document Frequency. It gives a higher importance score to words that are important in a particular document but not common in every document.

A word that appears frequently in one document can be useful, but a word that appears in almost every document may not help classification much. TF-IDF balances these two ideas.

Word Embeddings

Word embeddings represent words using dense numerical vectors. Unlike simple BoW, embeddings are designed so that words used in similar contexts can have similar numerical representations. This helps models capture more meaning.

Step 4: Model Training

After feature extraction, the numerical data is given to a machine-learning algorithm. The algorithm learns how the features are connected to the target categories.

- Naive Bayes: A probability-based classifier that is fast and commonly used for text.
- Logistic Regression: Learns how strongly different features are related to a class.
- Decision Tree: Makes decisions through a tree of conditions.
- Random Forest: Combines many decision trees to make a stronger prediction.
- Support Vector Machine (SVM): Tries to find a strong boundary between different classes.
- Neural Networks: Learn more complex patterns and can be useful for larger or more difficult NLP tasks.

Step 5: Prediction

Once trained, the model can classify new text that it has not seen before.

Prediction example
Input: "India won the football match."
Predicted category: Sports

Applications

- News categorization
- Email filtering
- Document organization
- Customer complaint routing
- Question classification

Advantages

- Fast classification of large amounts of text.
- Can achieve good accuracy with suitable data and features.
- Easy to automate once a model is trained.
- Reduces manual sorting work.

Limitations

- Usually needs correctly labeled training data.
- Ambiguous text can be difficult to classify.
- Noisy, incomplete, or poor-quality text can reduce performance.

2. Sentiment Analysis

Definition

Sentiment Analysis identifies the emotional tone, opinion, or attitude expressed in text. A basic sentiment system commonly classifies text as Positive, Negative, or Neutral.

Sentence	Sentiment	Reason
The movie was amazing.	Positive	"amazing" expresses a favorable opinion.
The food was terrible.	Negative	"terrible" expresses dissatisfaction.
The meeting starts at 5 PM.	Neutral	The sentence mainly states a fact.

How Sentiment Analysis Works

1. Customer Reviews / Text - Collect the text to be analyzed.
2. Preprocessing - Clean and prepare the text.
3. Feature Extraction - Convert the text into useful numerical representations.
4. ML Model - Learn the connection between language patterns and sentiment.
5. Output - Positive, Negative, or Neutral.

During training, the model sees many examples. It may learn that words such as excellent, amazing, and wonderful often appear in positive text, while terrible, worst, and disappointing often appear in negative text. More advanced models also consider context.

Algorithms Mentioned in the Notes

- Naive Bayes: Useful for simple and fast sentiment classification.
- Logistic Regression: A strong classical baseline for positive/negative classification.
- SVM: Often effective when text is represented using TF-IDF.
- Random Forest: Uses multiple decision trees.
- LSTM: A neural-network architecture designed to process sequences and remember useful earlier information.
- BERT: A transformer-based language model that can understand context more deeply than traditional word-frequency methods.

Applications

- Product review analysis
- Social media monitoring
- Brand reputation analysis
- Political opinion analysis

- Customer feedback analysis

Why Sentiment Analysis is Useful

Organizations can receive thousands or millions of comments. Reading every comment manually is slow. Sentiment analysis can quickly summarize how people feel and help identify common positive or negative reactions.

Advantages

- Helps understand customer opinion.
- Can support product or service improvement.
- Processes large volumes of feedback quickly.

Limitations

- Sarcasm: A sentence may use positive words but mean something negative.
- Mixed opinions: One review can contain both positive and negative views.
- Language variation: Slang, spelling variations, emojis, and different languages make analysis harder.

> Sarcasm example
> "Great, my phone crashed again." The word "Great" looks positive, but the full sentence expresses frustration.

3. Spam Detection

Definition

Spam Detection identifies messages that are unwanted, fraudulent, irrelevant, or suspicious. A common classification is Spam versus Ham, where Ham means a genuine or normal message.

Spam example from the notes Congratulations! You have won ₹10 lakh. Click here. Output: Spam
Normal email example Meeting scheduled tomorrow at 10 AM. Output: Not Spam / Ham

Workflow

1. Emails / Messages
2. Cleaning - Prepare the message text and useful metadata.
3. Feature Extraction - Convert important clues into numbers.
4. Classifier - Predict the class.
5. Output - Spam or Ham.

Common Features Used for Spam Detection

- Suspicious-word frequency: Repeated use of words connected with prizes, urgent offers, money, or suspicious requests may be useful clues.
- Number of links: Spam messages may contain many links or suspicious URLs.
- Special characters: Excessive symbols can be a signal.
- Sender information: The sender's identity or reputation can help.
- Capital letters: Excessive uppercase text may be one feature.

A good spam detector normally combines many features. One suspicious word alone should not automatically decide that a message is spam.

Algorithms

- Naive Bayes
- Decision Tree
- Random Forest
- SVM
- Logistic Regression

Applications

- Email services

- SMS filtering
- Social media filtering
- Fraud-related message detection

Advantages

- Protects users from unwanted or suspicious messages.
- Saves storage and reduces inbox clutter.
- Can help reduce exposure to phishing messages.

Limitations

- False positives: A genuine message may incorrectly be marked as spam.
- Changing techniques: Spammers constantly change wording and methods, so models may need new training data and updates.

Important term
False Positive: A normal message is incorrectly predicted as spam.
False Negative: A spam message is incorrectly allowed into the normal inbox.

4. Topic Modeling (LDA)

Definition

Topic Modeling automatically discovers hidden themes or topics in a collection of documents. Unlike classification, the documents do not need predefined category labels.

The notes introduce Latent Dirichlet Allocation (LDA) as a popular topic-modeling algorithm. LDA is an unsupervised learning technique.

Supervised vs Unsupervised - Easy Difference

- Supervised: The training examples have known answers or labels.
- Unsupervised: The system receives data without labels and tries to discover structure or patterns by itself.

Example
Documents containing football, cricket, team, and World Cup may form a Sports topic.
Documents containing bank, investment, money, and loan may form a Finance topic.

Main Idea Behind LDA

LDA assumes two important things:

- A document can contain multiple topics. For example, an article may be mostly about sports but may also mention business.
- Each topic is represented by multiple words. A Sports topic may strongly contain words such as cricket, match, player, team, and cup.

Topic probability example from the notes
Document: "The cricket team won the world cup."
Sports -> 90%
Politics -> 5%
Business -> 5%

These percentages express the idea that a document can be a mixture of topics instead of belonging to only one category.

How LDA Works - Conceptual Steps

1. Documents - Start with a collection of text documents.
2. Preprocessing - Clean, tokenize, and prepare the words.
3. Document-Term Matrix - Represent how words occur across documents.
4. LDA Algorithm - Discover groups of words that tend to occur as topic patterns.
5. Topics Generated - Inspect the important words and interpret each discovered topic.

Document-Term Matrix

A Document-Term Matrix is a table where rows represent documents and columns represent words. The values describe how often a word appears, or some related numerical importance.

It gives the algorithm a numerical representation of the document collection.

Interpreting an LDA Topic

LDA usually gives a group of high-probability words rather than a human-written topic name. A person often looks at those words and assigns a meaningful label.

> Example
> Top words: cricket, team, match, player, cup -> Human interpretation: Sports

Applications

- News grouping
- Research paper analysis
- Customer feedback analysis
- Legal document organization
- Healthcare record analysis

Advantages

- Does not require labeled data.
- Can reveal hidden patterns in large document collections.
- Helps organize large datasets.

Limitations

- Choosing the number of topics: The user often needs to decide how many topics the model should find.
- Interpretability: The generated word groups may not always form a clear human-readable topic.

5. Keyword Extraction

Definition

Keyword Extraction identifies the most important words or phrases in a document. Keywords summarize the main ideas of the text and make it easier to understand, search, index, or organize.

> Example from the notes
> Document: Artificial Intelligence is transforming healthcare through machine learning and NLP.
>
> Possible keywords: Artificial Intelligence, Healthcare, Machine Learning, NLP

How Keyword Extraction Works

1. Document
2. Cleaning - Remove unnecessary noise.
3. Tokenization - Split text into useful units.
4. Word Scoring - Give words or phrases importance scores.
5. Top Keywords - Select the highest-ranked items.

Methods

1. TF-IDF

TF-IDF can be used to identify words that are important in one document but less common across the full document collection. Words with higher useful TF-IDF scores can become keyword candidates.

2. RAKE

RAKE stands for Rapid Automatic Keyword Extraction. It looks for important words and phrases using patterns such as word frequency and how words occur together. It is useful because keywords are often phrases rather than single words.

3. TextRank

TextRank is a graph-based ranking method inspired by PageRank. Words or phrases are treated as connected items in a graph. Important words gain higher ranks based on their relationships with other words.

4. Machine Learning

A trained machine-learning model can also learn which words or phrases should be treated as keywords. Such a model may use features from the text and examples where the correct keywords are known.

Applications

- Search engines

- SEO
- Document summarization
- Research papers
- Content recommendation

Advantages

- Saves reading time by showing the main terms.
- Improves searching.
- Supports better document indexing and organization.

Limitations

- A method may miss an important phrase even when individual words seem common.
- Results depend strongly on preprocessing quality.

Keyword Extraction vs Text Classification

Keyword extraction finds important words or phrases inside a document. Text classification assigns the whole document to one or more predefined categories. They solve different problems.

6. Named Entity Recognition (NER)

Definition

Named Entity Recognition (NER) identifies important real-world entities in text and assigns them an entity type.

The notes list common entity categories such as:

- Person
- Organization
- Location
- Date
- Time
- Money
- Percentage

Example from the notes Sentence: "Rohit Sharma plays for India." Rohit Sharma -> Person India -> Location/Country
Another example from the notes Sentence: "Google opened a new office in Mumbai on Monday." Google -> Organization Mumbai -> Location Monday -> Date

How NER Works

1. Sentence - Input text is provided.
2. Preprocessing - The sentence is prepared for analysis.
3. Feature Extraction / Representation - The system represents useful language information.
4. NER Model - The model predicts entity types for relevant words or spans.
5. Entity Labels - Output such as Person, Organization, Location, Date, etc.

NER is not simply searching for capital letters. A useful NER model learns context. For example, the same word could represent a person, company, or location depending on the sentence.

Classical Machine-Learning Algorithms

- Hidden Markov Model (HMM): A probabilistic sequence model that can model a sequence of hidden labels.
- Conditional Random Fields (CRF): A sequence-labeling method that considers relationships between neighboring labels.

- Maximum Entropy Models: Probability-based models that can combine multiple features.

Deep-Learning Methods

- BiLSTM: Reads sequence information in both forward and backward directions, helping it use surrounding context.
- BiLSTM-CRF: Combines a BiLSTM's learned contextual features with CRF-based sequence labeling.
- Transformer Models: Modern models that use attention to capture relationships between words.
- BERT: A transformer-based language model that produces context-aware representations.
- RoBERTa: A transformer model related to BERT with modified training choices.

Applications

- Search engines
- Chatbots
- Resume parsing
- Medical records
- Question answering
- Information extraction

Advantages

- Automatically extracts useful structured information from text.
- Reduces manual data-entry and reading effort.
- Can improve search and information retrieval.

Limitations

- Ambiguous names: A name may have more than one possible entity type.
- Domain-specific entities: Specialized fields may require custom training.
- Multiple meanings: Context is needed when words or names have different meanings.

Why context matters
"Apple released a new device" -> Apple is likely an Organization.
"She ate an apple" -> apple is a common noun, not a company entity.

Comparison of Classical NLP Tasks

Task	Learning Type	Input	Output	Common Methods	Applications
Text Classification	Supervised	Documents	Categories	Naive Bayes, SVM, Logistic Regression	News classification, email sorting
Sentiment Analysis	Supervised	Reviews, tweets	Positive / Negative / Neutral	Naive Bayes, SVM, BERT	Product reviews, social media
Spam Detection	Supervised	Emails, SMS	Spam / Ham	Naive Bayes, Random Forest	Email filtering
Topic Modeling (LDA)	Unsupervised	Documents	Hidden topics	LDA	Research analysis, news grouping
Keyword Extraction	Unsupervised / Statistical / ML	Documents	Important keywords	TF-IDF, RAKE, TextRank	SEO, search engines
Named Entity Recognition	Supervised	Sentences	Named entities	CRF, BiLSTM, BERT	Chatbots, search, information extraction

How to Remember the Difference

- Text Classification: What category does this text belong to?
- Sentiment Analysis: What opinion or emotion does this text express?
- Spam Detection: Is this message spam or genuine?
- Topic Modeling: What hidden themes exist in these documents?
- Keyword Extraction: What are the most important words or phrases?
- NER: Which words refer to people, organizations, places, dates, and other entities?

Supervised vs Unsupervised Tasks in This Chapter

Supervised: Text Classification, Sentiment Analysis, Spam Detection, and the NER framing in the college comparison table use labeled examples. Unsupervised: Topic Modeling (LDA) discovers topics without labels. Keyword Extraction can use statistical, unsupervised, or machine-learning approaches depending on the method.

Quick Exam Revision

One-line Definitions

- NLP: AI field that enables computers to understand, interpret, generate, and respond to human language.
- Text Classification: Automatically assigns predefined labels to text.
- Sentiment Analysis: Identifies whether an opinion is positive, negative, or neutral.
- Spam Detection: Identifies unwanted or fraudulent messages and separates Spam from Ham.
- Topic Modeling: Discovers hidden topics from unlabeled document collections.
- LDA: An unsupervised topic-modeling technique in which documents are mixtures of topics and topics are mixtures of words.
- Keyword Extraction: Finds the most important words or phrases in a document.
- NER: Finds and classifies named entities such as people, organizations, locations, dates, time, money, and percentages.

Important Full Forms

Short Form	Full Form
NLP	Natural Language Processing
AI	Artificial Intelligence
ML	Machine Learning
BoW	Bag of Words
TF-IDF	Term Frequency-Inverse Document Frequency
SVM	Support Vector Machine
LDA	Latent Dirichlet Allocation
RAKE	Rapid Automatic Keyword Extraction
NER	Named Entity Recognition
HMM	Hidden Markov Model
CRF	Conditional Random Fields
LSTM	Long Short-Term Memory
BiLSTM	Bidirectional Long Short-Term Memory

Typical Answer Structure for a 5- or 6-Mark Question

For any major NLP task, a clear exam answer can be written in this order:

- 1. Definition
- 2. Main purpose
- 3. Workflow / steps

- 4. Algorithms or methods
- 5. One simple example
- 6. Applications
- 7. Advantages
- 8. Limitations

This structure matches the way the topics are organized in the college notes and makes answers easy to revise.

Final Concept Map

Human Text
Clean / Preprocess
Convert into Features or Contextual Representations
Apply Statistical / ML / Deep-Learning Method
Produce an NLP Output: category, sentiment, spam label, topic, keyword, or entity

End of Unit 2 - Chapter 1: Machine Learning Approaches in NLP